

SYSTEMS ANALYSIS

Systems Analysis & Design

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2025-III



Outline

- 1 Basic Concepts
- 2 Chaos and Dynamic Systems
- 3 Abstraction and Modularity



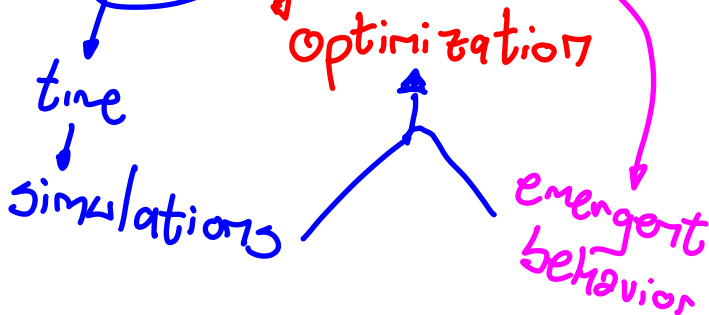
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What is a System?

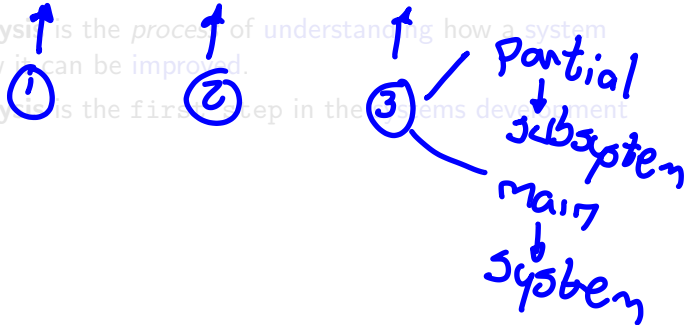
- A **system** is a set of interacting components that work together to achieve a common goal.
- A **system** is a collection of elements organized in a specific way.
- A **system** is a structure designed to perform a specific function.



Systems Analysis Process

- **Systems analysis** is the *process* of *studying* a *system* in order to identify its *components*, *interactions*, and *goals*.

- Systems analysis is the *process* of understanding how a system works and how it can be improved.
- Systems analysis is the first step in the systems development lifecycle.



Systems Analysis Process

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- **Systems analysis** is the *first* step in the *systems development* lifecycle.

manipulate



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Requirements



Systems Development Lifecycle – SDLC

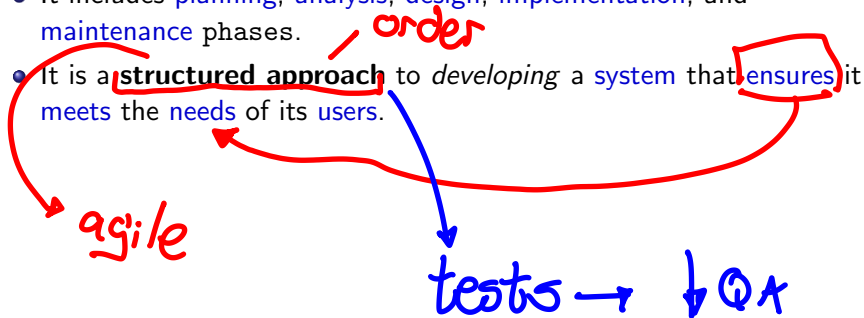
- The **systems development lifecycle** is a **process** that **guides** the development of a system **→ software**
- It includes **planning**, **analysis**, design, implementation, and maintenance phases.
- It is a **structured approach** to developing a system that ensures it meets the needs of its users.

architecture



Systems Development Lifecycle

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Systems Analysis Techniques

- **Systems analysis** uses a variety of techniques to study a system.
- These include interviews, surveys, observations and document analysis.

- It also includes data modeling, process modeling, and requirements analysis.

open
dynaical

sponsors
final users

closed

Netflix

gather
information

BlackBuster \$1 (\$3)



①



fixing



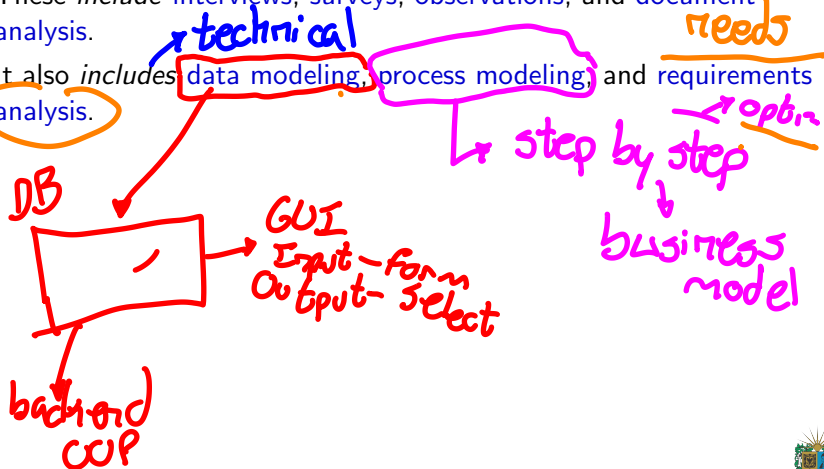
1997

no punishment



Systems Analysis Techniques

- **Systems analysis** uses a *variety* of **techniques** to *study* a **system**.
- These *include* interviews, surveys, observations, and document analysis.
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Systems Analysis Tools

- **Systems analysis** uses a variety of tools to study a system.
- These include diagrams, charts, flowcharts, and data models.
- It also includes software tools such as spreadsheets, databases, and simulation software.

architecture
system

plots

descriptive
analysis

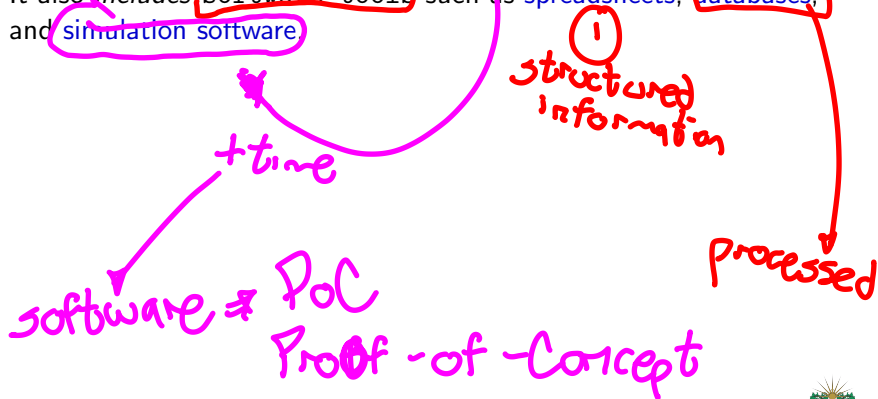
Time-series Columns Boxplot

each
steps-process



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Lateral Thinking

- **Lateral thinking** is a creative problem-solving technique that involves thinking outside the box.
- It is a non-linear approach to problem-solving that encourages innovation and creativity.
- It is a useful technique for generating new ideas and solving complex problems.

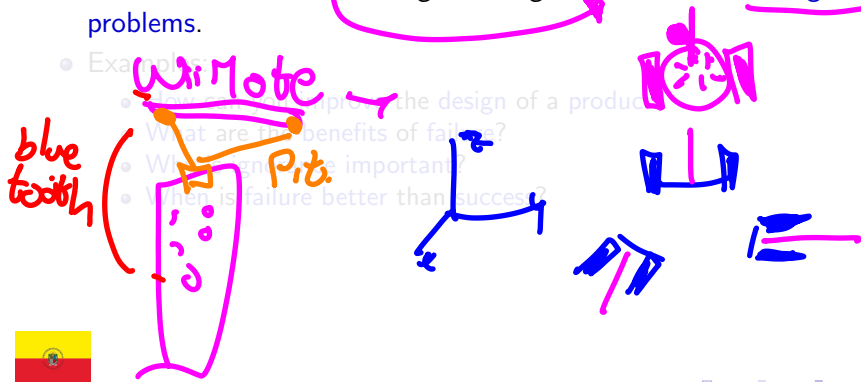
• Examples:

• How can you improve the design of a product?

• What are the benefits of failure?

• What design is important?

• When is failure better than success?



- How can you improve the design of a product?
- What are the benefits of failure?
- Why is ~~ignorance~~ important? *learn*
- When is failure better than success?

business opportunity

Steve Jobs

AWZ



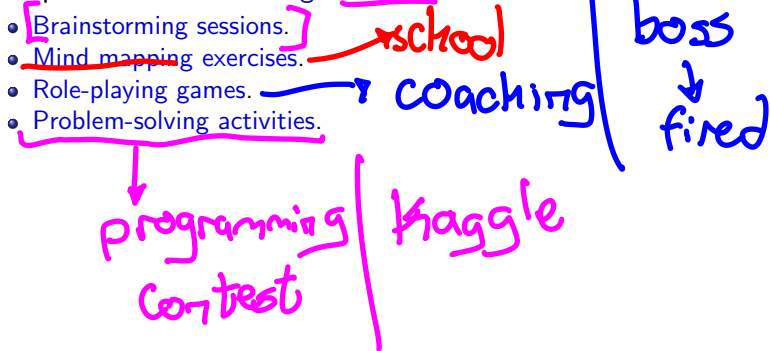
Lateral Thinking Training

- **Lateral thinking** is a *skill* that can be *learned* and *developed* through training and practice. *Go/chess*
- It involves exercises, games, and activities that encourage creative thinking.
- Examples of lateral thinking exercises:
 - Brainstorming sessions.
 - Mind mapping exercises.
 - Role-playing games.
 - Problem-solving activities.



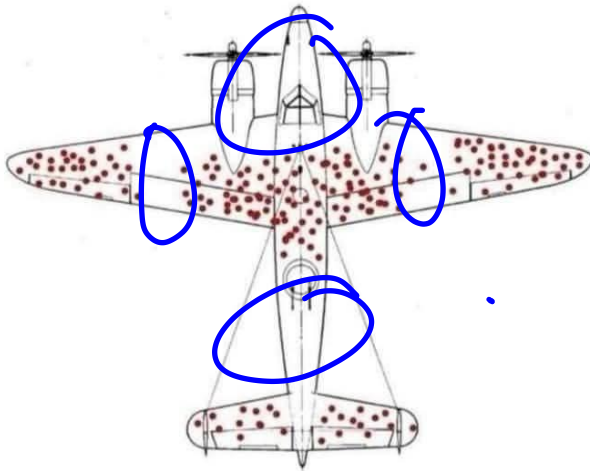
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- Examples of lateral thinking exercises:



Lateral Thinking Exercise

World War II, the perspective of Abraham Wald:



Uncertainty and Risk

- **Uncertainty** is the lack of knowledge about the **future outcome** of a decision or event.
 - Risk is the probability of a negative outcome or loss associated with a decision or event.
 - Uncertainty and risk are *inherent* in complex systems and decisions.
 - They can be managed through planning, analysis, and mitigation strategies.
- Handwritten notes:*
A blue arrow points from "future outcome" to "Uncertainty".
Two blue arrows point from "no prediction" to "Risk".



Uncertainty and Risk

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- Handwritten note:* identify → characterize



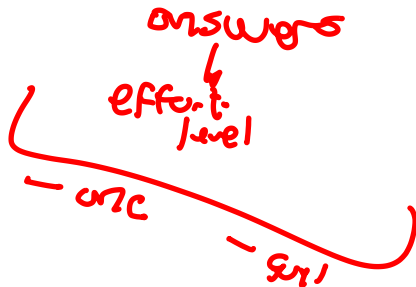
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Complexity and Emergence

- **Complexity** is the degree to which a system is difficult to understand.
- **Emergence** is the appearance of unexpected properties in a system that arise from the interactions of its components.
- **Complexity and emergence** are common in dynamic systems that are non-linear and chaotic.
- They can be studied and understood through systems analysis and modeling.

Planet Nine



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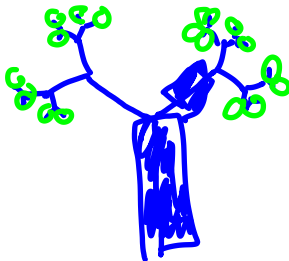
Lodenmeyer-System

1 Basic Concepts

2 Chaos and Dynamic Systems

3 Abstraction and Modularity

$$D \rightarrow L(D/2) + L(D \# 2)$$



What is Chaos?

- **Chaos** is a branch of mathematics that studies the sensitivity of dynamical systems to [initial conditions].
- **Chaos** is a non-linear behavior that is highly sensitive to initial conditions.

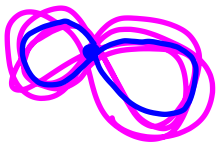
- Chaos is a deterministic behavior that is not predictable in the long term.
- Chaos is a complex behavior that is hard to understand.

decision making
 simulation → play with inputs
 data → kaggle
 context
 - tDA → ydata-profiling
 - Correlation Matrix
 A → B
 C

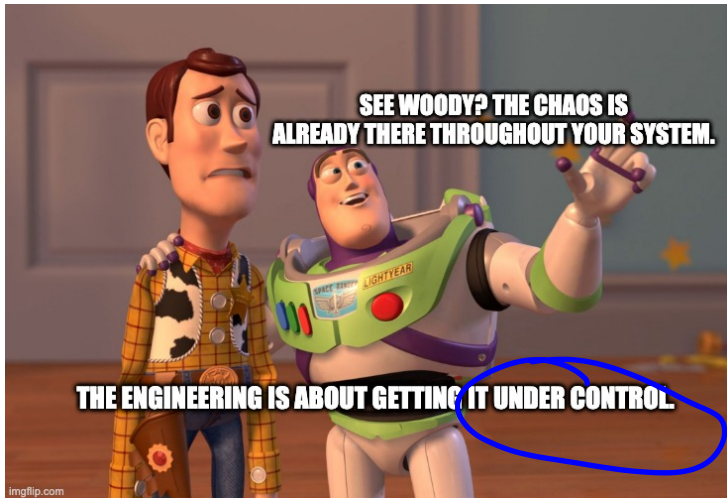


What is Chaos?

- **Chaos** is a *branch* of **mathematics** that *studies* the **sensitivity** of dynamical systems to **initial conditions**.
- **Chaos** is a **non-linear** behavior that is **highly sensitive** to initial conditions.
- **Chaos** is a **deterministic** behavior that is **not predictable** in the long term.
- **Chaos** is a **complex** behavior that is **hard** to understand.



Chaos is Everywhere!



What is a Dynamic System?

- A dynamic system is a system that changes over time.
- A dynamic system is sensitive to initial conditions.

- A dynamic system is non-linear.

- A dynamic system can be chaotic.

Chaotic

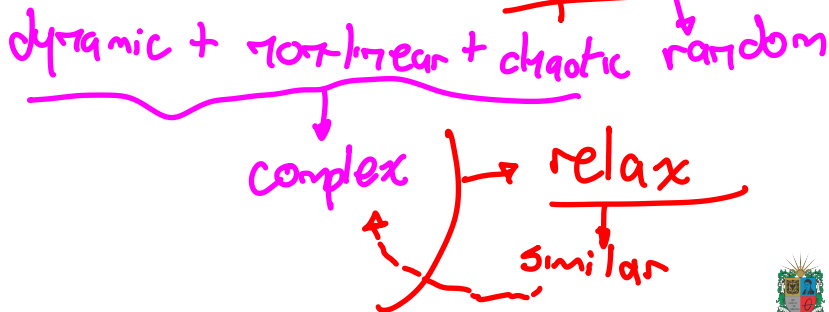
structure
+
time
dimension
simulation

inputs
+
sensitivity
control



What is a Dynamic System?

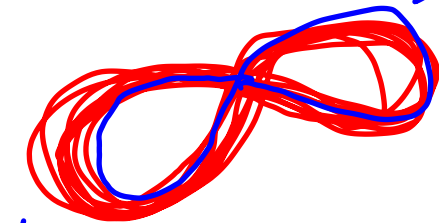
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- A **dynamic system** is **non-linear**.
- A **dynamic system** can be **chaotic**.



Chaotic Attractors

A chaotic attractor is a set of points in a phase space that attracts the trajectories of a dynamical system.

Lorenz

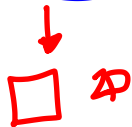


pattern
↓
behavior

no pattern → disorder



Fractals in Nature



Fibonacci

L-system



fractional

2.1 D
1.8 D
1.7 D
2.4 D

Watch this video: <https://www.youtube.com/watch?v=kkGeOWYOFoA>



Swarm Intelligence I

- **Swarm intelligence** is the collective behavior of decentralized, self-organized systems, natural or artificial.

- The concept is employed in work on artificial intelligence.

- The expression was introduced by Gerardo Beni and Jing Wang in 1989, in the context of cellular robotic systems. For example, watch this video.

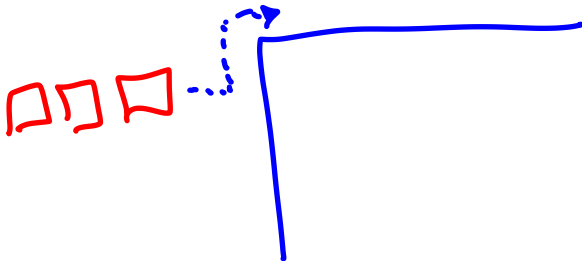
individual ~ not smart
but
population ~ smart

low-resources
knowledge
emergent
behaviors



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Swarm Intelligence II

Useless

- The idea is: if you see an individual, it may seem random; however several individuals interacting with each other and the environment show smart behaviors.
- Yu Takeuchi said: one colombian is more intelligent than one japanese, but two colombians are smarter than two colombians.
- There are interesting **population behaviors** in nature, specially in insects: bees, ants, termites, among others.
- There are also many examples in nature: schools of fish, birds, wolves.

emergent



Swarm Intelligence II

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maths
to Colombia for no team



Swarm Intelligence II

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bio-inspired computation

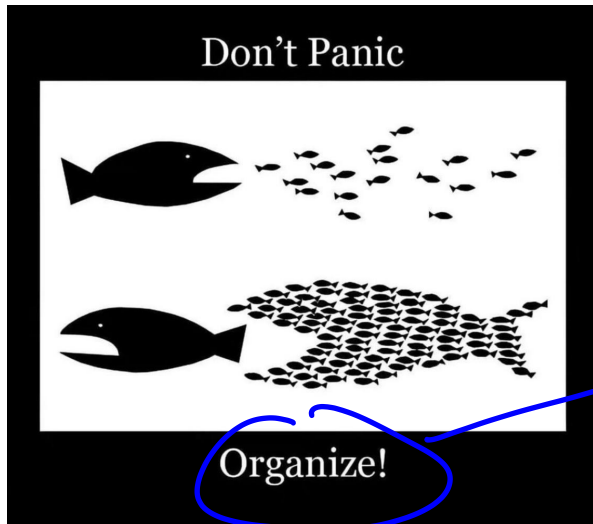


Emergent Behaviors

- Emergent behavior is the appearance of complex patterns and behaviors from many simple interactions. *→ dynamical system*
- Emergent behavior **results** from the collective behavior of the individuals in the system. *→ multi-agent system*
- Emergent behavior is not planned or designed by any individual, but **arises** from their interactions.
- Emergent behavior is not the sum of the **individual** behaviors, but something more: synergy.
- Swarm intelligence** refers to interesting emergent behaviors.



School Fish Algorithm



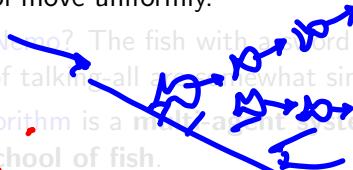
School Fish Algorithm

- **School fish** are interesting. When a predator attacks, they become confused by the large number of individuals and their diverse movements.
- The idea is simple: *"Don't touch me, don't come too close, but stay somewhat close."*
- This behavior is a chain of action and reaction. It confuses predators and helps the school move uniformly.
- Do you remember Nemo? The fish with a sword snout, the pirates, or Marlin's imitation of talking-all are somewhat similar. Watch **here**.
- The school fish algorithm is a multi-agent system that simulates the behavior of a school of fish.



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Ant Colony Algorithm

- Ant colony algorithm is a **multi-agent system** that **simulates** the behavior of an **ant colony**.
- Ant colony algorithm is based on the **social behavior** of **ants** and the use of **pheromones**. Watch [here](#).
- Ant colony algorithm is used to **solve optimization problems**. Watch [here](#).



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What is Abstraction?

- **Abstraction** is the *process* of **ignoring** minor **details** in order to **focus** on the **important aspects** of a **system**.
- **Abstraction** is the *process* of **simplifying** a complex system to **understand** it.
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- **Modularity** is the *process* of **organizing** a **system** into **independent units** that can be **developed** and **maintained** separately.
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Abstraction and Modularity

- **Abstraction** and **modularity** are *two important concepts* in **systems analysis**.
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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

